

Sedrick Scott Keh

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EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA
MS in Machine Learning, GPA: 4.11/4.33 *Sept. 2021 – Dec. 2022*
- **Hong Kong University of Science and Technology** Hong Kong
BSc in Data Science; BSc in Mathematics – First Class Honors, GPA: 4.09/4.30 *Sept. 2017 – Jun. 2021*

PEER-REVIEWED PUBLICATIONS

- [1] [PINEAPPLE: Personifying INanimate Entities by Acquiring Parallel Personification data for Learning Enhanced generation](#)
***Sedrick Scott Keh**, Kevin Lu, Varun Gangal, Steven Y. Feng, Harsh Jhamtani, Malihe Alikhani and Eduard Hovy.*
International Conference on Computational Linguistics (COLING 2022). October 2022.
- [2] [EUREKA: EUPhemism Recognition Enhanced through Knn-based methods and Augmentation](#)
Sedrick Scott Keh, Rohit K. Bharadwaj*, Emmy Liu, Simone Tedeschi, Varun Gangal and Roberto Navigli.*
The Third Workshop on Figurative Language Processing (EMNLP 2022). December 2022.
- [3] [Exploring Euphemism Detection in Few-Shot and Zero-Shot Settings](#)
Sedrick Scott Keh.
The Third Workshop on Figurative Language Processing (EMNLP 2022). December 2022.
- [4] [NewsPanda: Media Monitoring for Timely Conservation Action](#)
Sedrick Scott Keh, Zheyuan Ryan Shi*, David J. Patterson, Nirmal Bhagabati, Karun Dewan, Areendran Gopala, Pablo Izquierdo, Debojyoti Mallick, Ambika Sharma, Pooja Shrestha, and Fei Fang.*
Conference on Innovative Applications of Artificial Intelligence (IAAI 2023). February 2023.

PREPRINTS AND PAPERS IN SUBMISSION

- [5] [PANCETTA: Phoneme Aware Neural Completion to Elicit Tongue Twisters Automatically](#)
***Sedrick Scott Keh**, Steven Y. Feng, Varun Gangal, Malihe Alikhani and Eduard Hovy.*
arXiv:2209.06275. September 2022. (Under review at EACL 2023)
- [6] [Hashtag-Guided Low-Resource Tweet Classification](#)
Shizhe Diao, **Sedrick Scott Keh***, Liangming Pan, Zhiliang Tian, Yan Song and Tong Zhang.*
October 2022. (Under review at WWW 2023)
- [7] [Semi-Supervised Noisy Student Pre-training on EfficientNet Architectures for Plant Pathology Classification](#)
Sedrick Scott Keh
arXiv:2012.00332. December 2020.
- [8] [Myers-Briggs Personality Indicator Classification and Language Generation Using Pre-trained Language Models](#)
***Sedrick Scott Keh**, I-Tsun Cheng.*
arXiv:1907.06333. July 2019.
- [9] [Crystallographic Lattice Classification with Deep Learning and MagmaDNN](#)
***Sedrick Scott Keh**, Daniel Nichols, Kam Fai Chan.*
Technical Report. August 2019.
- [10] [MagmaDNN: Towards High-Performance Deep Learning Using Magma](#)
*Daniel Nichols, **Sedrick Scott Keh**, Kam Fai Chan.*
Technical Report. July 2019.

RESEARCH EXPERIENCE

- **Carnegie Mellon University – Language Technologies Institute** Sept. 2021 - Present
Advised by Prof. Daniel Fried and Prof. Eduard Hovy
 - **Language Grounding:** Working on asking informative clarifying questions in a collaborative and visually-grounded uncertainty resolution task. Paper in progress.
 - **Creative Text Generation:** Using commonsense reasoning to generate figurative language.
 1. Proposed a model to generate diverse and meaningful personifications by incorporating commonsense concepts like animacy. First-author paper accepted at **COLING 2022** [1].
 2. Designed a system to automatically detect euphemisms in a sentence. System won **1st place** at the Euphemism Detection Shared Task at the FigLang Workshop at EMNLP 2022. Also investigated euphemism detection in a few-shot and zero-shot setting. Two papers (1 first-author, 1 solo-author) accepted at **EMNLP 2022 FigLang Workshop** [2] [3].
 - **Controllable Text Generation:** Gathered a dataset of tongue twisters and proposed a model to generate tongue twisters by modeling linguistic difficulty. First-author paper under review [5].
- **Carnegie Mellon University – Institute for Software Research** Sept. 2021 - Present
Advised by Prof. Fei Fang
 - **AI/NLP for Social Good:** Deployed a system to automatically extract and classify weekly news articles related to wildlife conservation. This is a collaboration with WWF offices in UK, India, and Nepal. First-author paper accepted to **IAAI 2023** [4].
- **HKUST Statistics and Machine Learning Group** Feb. 2020 - Dec. 2021
Advised by Prof. Tong Zhang
 - **Social Media Text Generation:** Created an intelligent social media captioning model which makes predictions by considering closely related posts and replies. Co-first-author paper under review [6].
 - **Story Generation:** Explored prompt-based methods for story completion tasks.

AWARDS AND SCHOLARSHIPS

- **Rank 1** – EMNLP 2022 Figurative Language Workshop Shared Task 2022
- **Scholarship** – HKUST Full Undergraduate Scholarship (\$560,000 HKD) 2017-2021
- **Scholarship** – S.S. Chern Class Mathematics Achievement Scholarship (\$30,000 HKD) 2018-2021
- **Finalist (Top 20; training pool for IMO)** – Philippine Mathematical Olympiad 2015, 2016, 2017
- **Finalist (Top 30; training pool for IOI)** – National Olympiad in Informatics 2017

ACADEMIC SERVICE AND TEACHING

- **Program Committee:** ACL 2023
- **Reviewer:** EMNLP 2022, FigLang Workshop at EMNLP 2022
- **Conference Volunteer:** EMNLP 2022 (Abu Dhabi, UAE)
- **Teaching Assistant:** 10-617 Intermediate Deep Learning (Graduate-level) Fall 2022
- **Teaching Assistant:** 10-315 Advanced Introduction to Machine Learning Spring 2022
 - Nominated for the 2022 Alan J. Perlis Graduate Student Teaching Award
- **Mentor:** CMU Undergraduate AI Mentorship Program

INDUSTRY EXPERIENCE

- **Thinking Machines Data Science** Jun. 2021 - Aug. 2021
Machine Learning Research Intern (Geospatial Team)
- **Senti AI Tech Labs** Jul. 2020 - Sept. 2020
NLP Research Intern